

# Mathematics—Grade 10 Quadratics with Exercises

**Subject:** Mathematics

A quadratic expression contains a variable raised to the power of two and can be written as  $ax^2 + bx + c$ , where  $a$  is not zero. A quadratic equation is obtained when this expression is set equal to a value, commonly zero. Quadratic equations can be solved by factoring, completing the square, or using the quadratic formula. Factoring is often the quickest method when the numbers are simple. For  $x^2 - 5x + 6 = 0$ , the numbers  $-2$  and  $-3$  multiply to  $6$  and add to  $-5$ , so the equation becomes  $(x - 2)(x - 3) = 0$ . The solutions are  $x = 2$  and  $x = 3$ .

The zero-product rule says that if two factors multiply to zero, at least one factor must be zero. For an equation such as  $(2x + 1)(x - 4) = 0$ , either  $2x + 1 = 0$  or  $x - 4 = 0$ . This gives  $x = -1/2$  or  $x = 4$ . Some quadratics do not factor easily. For  $ax^2 + bx + c = 0$ , the quadratic formula is  $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ . The expression  $b^2 - 4ac$  is called the discriminant. A positive discriminant gives two distinct real solutions, a zero discriminant gives one repeated real solution, and a negative discriminant gives no real solutions.

A quadratic function has the form  $y = ax^2 + bx + c$  and its graph is a parabola. If  $a$  is positive, the parabola opens upward and has a minimum point. If  $a$  is negative, it opens downward and has a maximum point. The axis of symmetry is  $x = -b/(2a)$ . The  $y$ -intercept is  $c$ , because setting  $x$  equal to zero gives  $y = c$ . The  $x$ -intercepts are the real solutions of the corresponding quadratic equation. In word problems, solutions must be checked in context. A negative length or negative time may be rejected even if it is an algebraic solution.

Exercise 1: Solve  $x^2 - 7x + 12 = 0$  by factoring.

Exercise 2: Solve  $2x^2 + 5x - 3 = 0$  by factoring.

Exercise 3: Solve  $4x^2 + 4x + 1 = 0$  by factoring.